Exercicio 1. Instalación de Apache:

- Captura 1 - Ficheiro de configuración de Vagrant (paso 2).

```
Vagrant.configure("2") do |config|
  config.vm.box = "debian/bookworm64"

  maquinas = [
    { vb_name: "prod-mtv", hostname: "prod-mtv", ip: "192.168.56.141" }
  ]

  maquinas.each do |m|
    config.vm.define m[:hostname] do |node|
      node.vm.hostname = m[:hostname]

      # Host-only network
      node.vm.network "private_network", ip: m[:ip]

      node.vm.provider "virtualbox" do |vb|
        vb.name = m[:vb_name]
        vb.memory = 1024
        vb.cpus = 1
      end

      node.vm.provision "shell", inline: <<-SHELL
        apt-get update
        apt-get install -y sudo openssh-server git

        # Crear usuario dadmin se non existe
        if ! id dadmin >/dev/null 2>&1; then
          useradd -m -s /bin/bash dadmin
          echo "dadmin:abc123." | chpasswd
        fi

        # Engadir a sudo
        usermod -aG sudo dadmin

        # Sudo sen contrasinal
        echo "dadmin ALL=(ALL) NOPASSWD:ALL" > /etc/sudoers.d/dadmin
        chmod 440 /etc/sudoers.d/dadmin

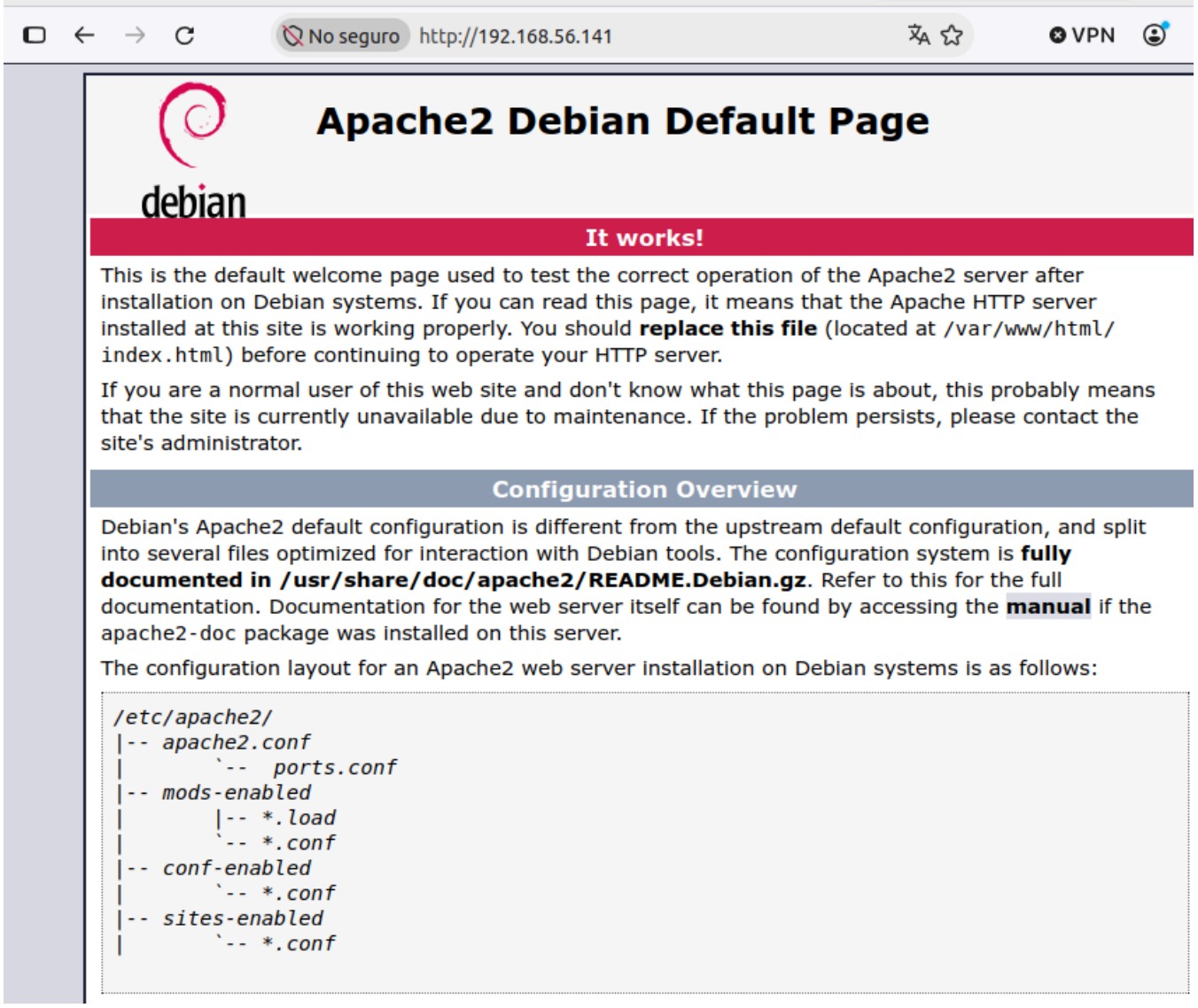
        # Permitir que se poida conectar mediante contrasinal
        sed -i 's/^#PasswordAuthentication no/PasswordAuthentication yes/; s/
        ^PasswordAuthentication no/PasswordAuthentication yes/' /etc/ssh/
        sshd_config

        # Activar SSH
        systemctl enable ssh
        systemctl restart ssh
      SHELL
    end
```

- Captura 2 - Comprobación da versión de Apache instalada (paso 7).

```
dadmin@prod-mtv:~$ sudo apache2ctl -v
Server version: Apache/2.4.68 (Debian)
Server built:   2026-06-12T05:08:45
dadmin@prod-mtv:~$
```

- Captura 3 - Páxina por defecto de Apache no navegador (paso 8).

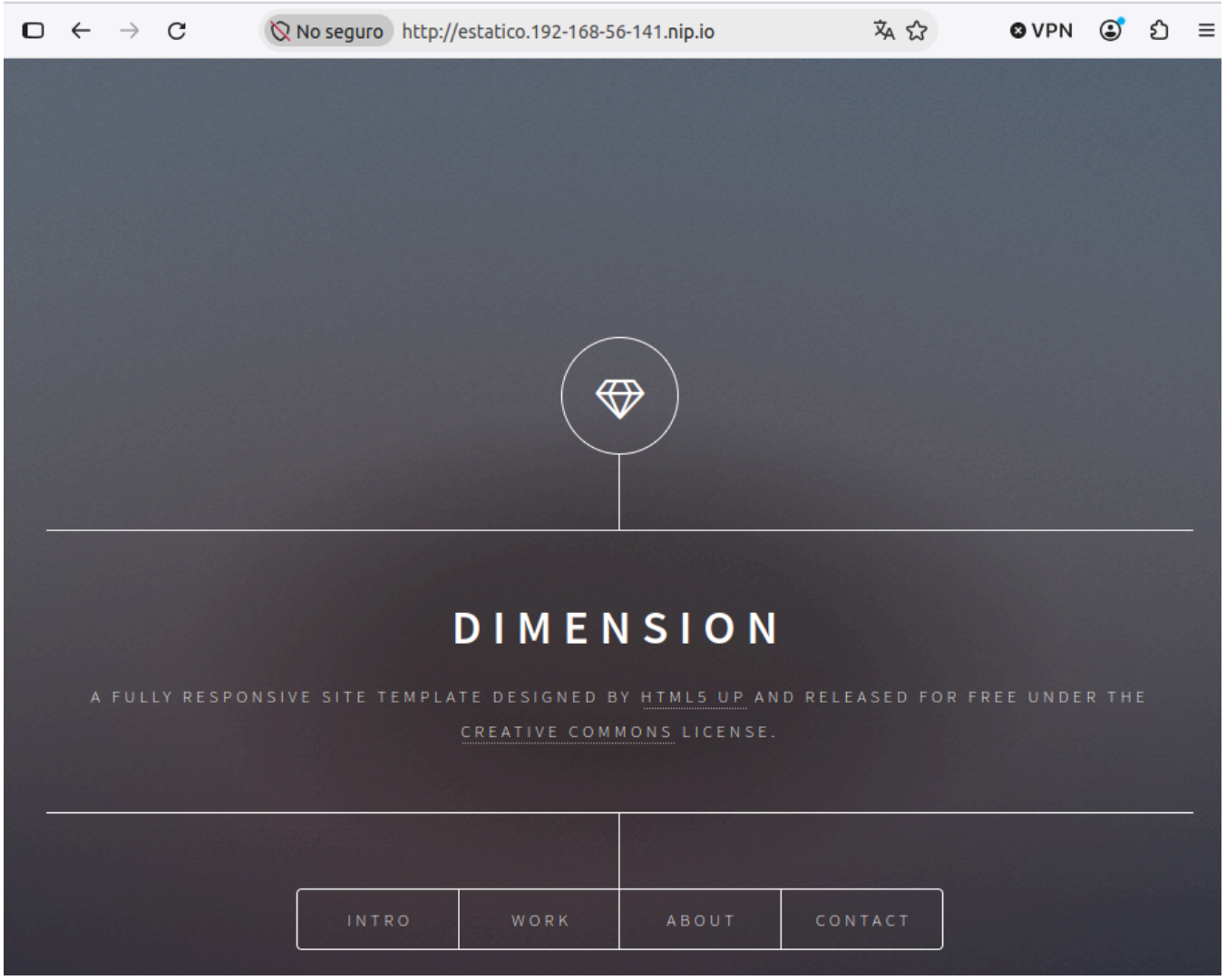


Exercicio 2. Creación de virtualhost:

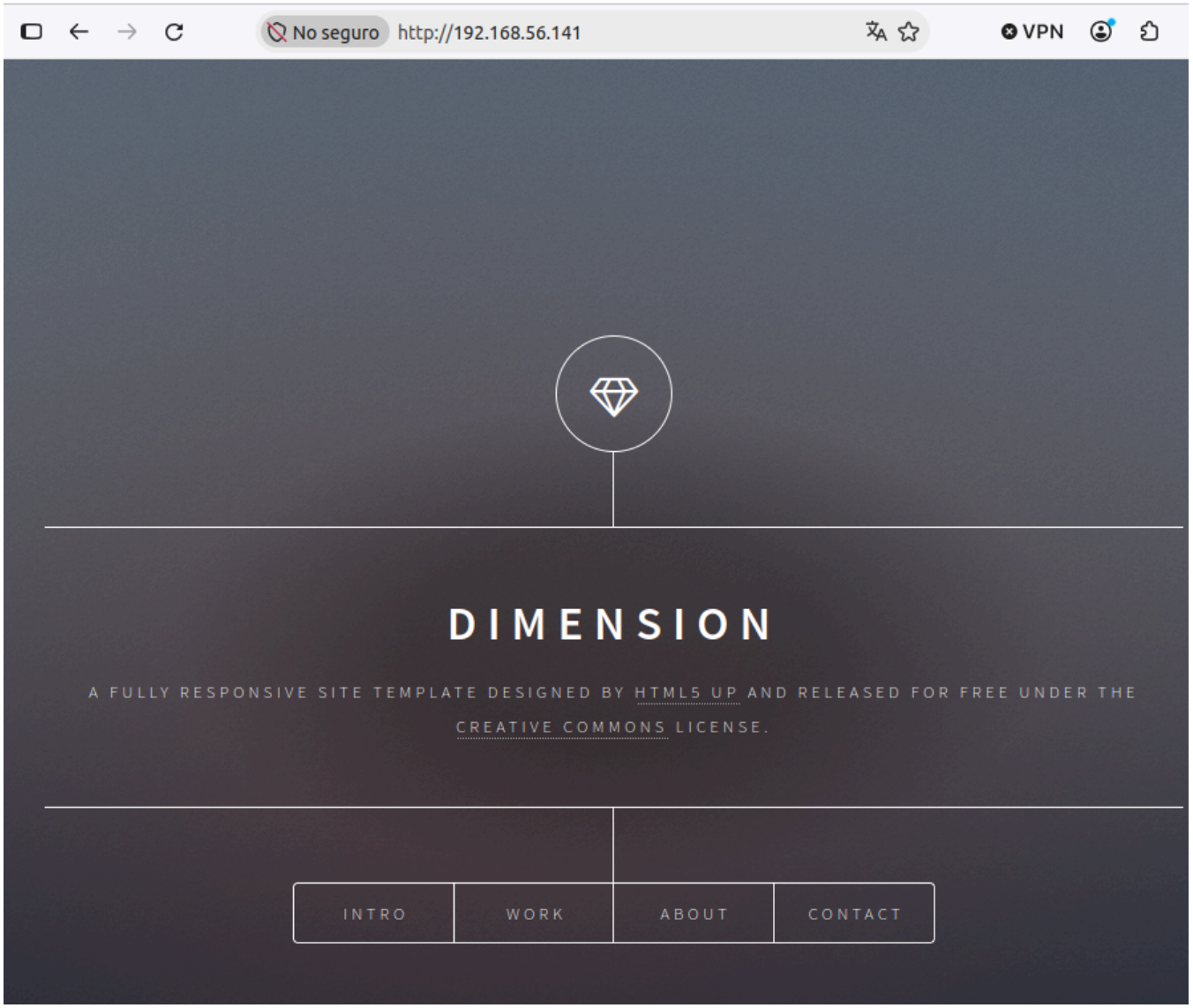
- Captura 1 - Definición do `virtualhost` (paso 7).

```
dadmin@prod-mtv:~$ sudo a2ensite estatico.conf
sudo apache2ctl configtest
sudo systemctl reload apache2
cat /etc/apache2/sites-available/estatico.conf
Enabling site estatico.
To activate the new configuration, you need to run:
  systemctl reload apache2
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Se
t the 'ServerName' directive globally to suppress this message
Syntax OK
<VirtualHost *:80>
    ServerName estatico.192-168-56-141.nip.io
    DocumentRoot /var/www/estatico/
</VirtualHost>
dadmin@prod-mtv:~$
```

- Captura 2 - Comprobación no navegador do funcionamento da web por nome `estatico` (paso 8).



- Captura 3 - Comprobación no navegador do funcionamento da web por IP como `virtualhost` por defecto (paso 10).

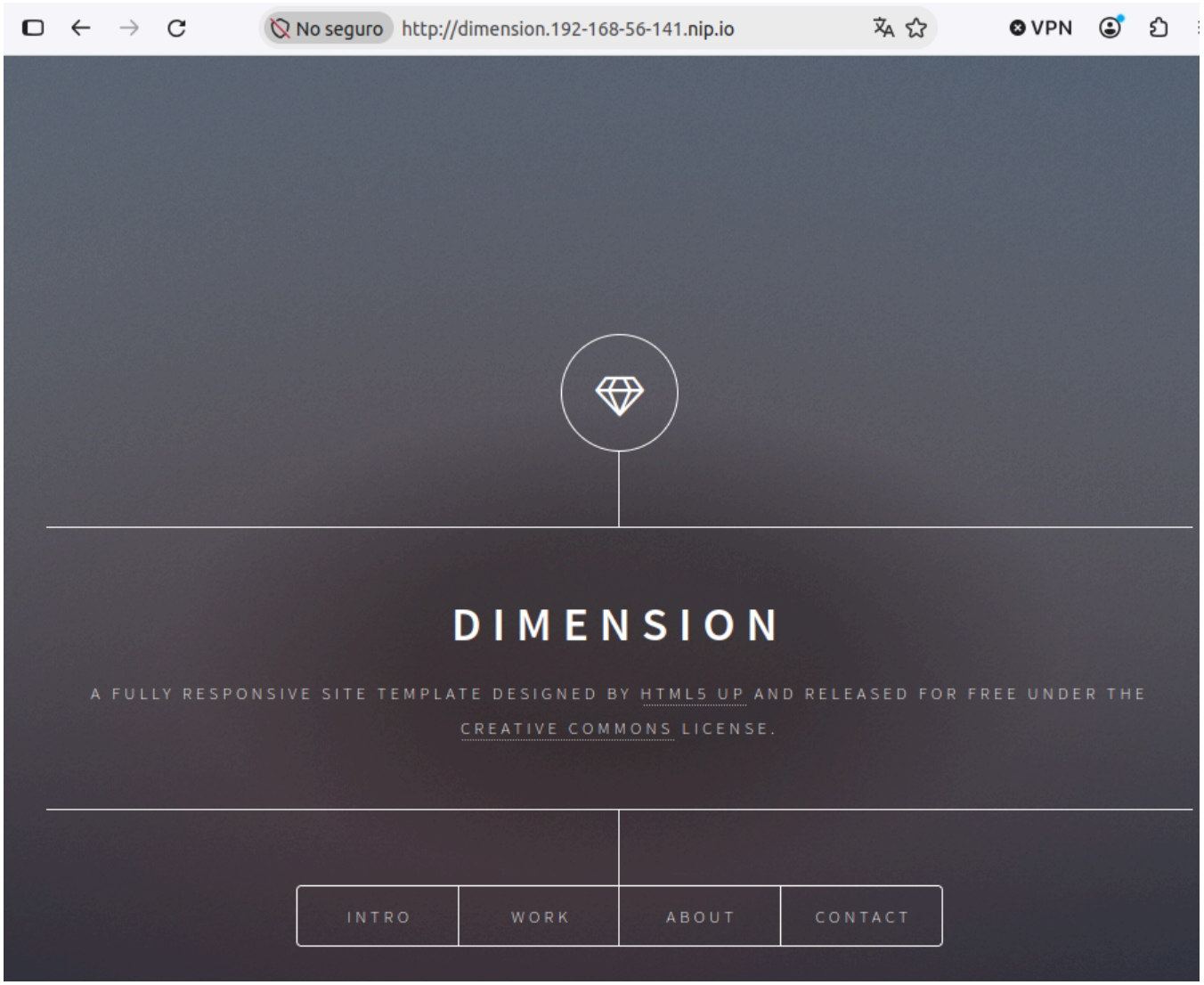


Exercicio 3. Directivas máis utilizadas:

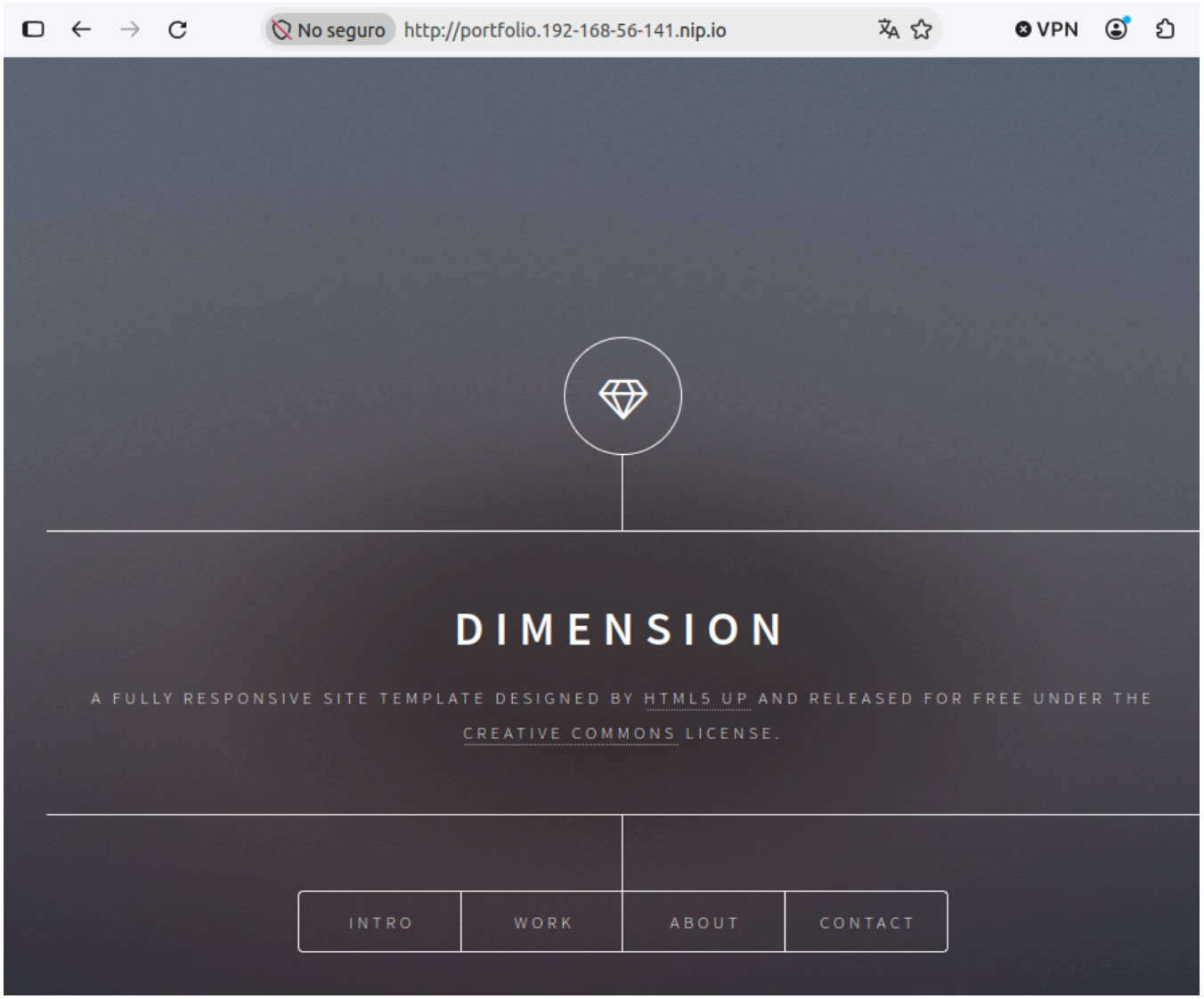
- Captura 1 - Definición do `virtualhost` cos alias (paso 1).

```
dadmin@prod-mtv:~$ sudo nano /etc/apache2/sites-available/estatico.conf
dadmin@prod-mtv:~$ sudo apache2ctl configtest
sudo systemctl reload apache2
cat /etc/apache2/sites-available/estatico.conf
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Se
t the 'ServerName' directive globally to suppress this message
Syntax OK
<VirtualHost *:80>
    ServerName estatico.192-168-56-141.nip.io
    ServerAlias dimension.192-168-56-141.nip.io portfolio.192-168-56-141.nip.io
    DocumentRoot /var/www/estatico/
</VirtualHost>
dadmin@prod-mtv:~$
```

- Captura 2 - Comprobación no navegador do funcionamento da web por nome `dimension` (paso 3).



- Captura 3 - Comprobación no navegador do funcionamento da web por nome `portfolio` (paso 3).



- Captura 4 - Definición do `virtualhost` cos *logs* (paso 5).

```
dadmin@prod-mtv:~$ sudo nano /etc/apache2/sites-available/estatico.conf
dadmin@prod-mtv:~$ sudo apache2ctl configtest
sudo systemctl reload apache2
cat /etc/apache2/sites-available/estatico.conf
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Se
t the 'ServerName' directive globally to suppress this message
Syntax OK
<VirtualHost *:80>
    ServerName estatico.192-168-56-141.nip.io
    ServerAlias dimension.192-168-56-141.nip.io portfolio.192-168-56-141.nip.io
    DocumentRoot /var/www/estatico/
    CustomLog /var/log/apache2/estatico-access.log combined
    ErrorLog /var/log/apache2/estatico-error.log
</VirtualHost>
dadmin@prod-mtv:~$
```

- Captura 5 - Contido do *log* de acceso (paso 8).

```
dadmin@prod-mtv:~$ sudo head -5 /var/log/apache2/estatico-access.log
192.168.56.1 - - [03/Nov/2026:10:55:38 +0000] "GET / HTTP/1.1" 200 3745 "-" "Mozilla/5.0 (X11; Linux x86_64;
 rv:159.0) Gecko/20100101 Firefox/159.0"
192.168.56.1 - - [03/Nov/2026:10:56:28 +0000] "GET / HTTP/1.1" 200 3745 "-" "Mozilla/5.0 (X11; Linux x86_64;
 rv:159.0) Gecko/20100101 Firefox/159.0"
192.168.56.1 - - [03/Nov/2026:10:56:28 +0000] "GET /assets/js/jquery.min.js HTTP/1.1" 200 33626 "http://esta
tico.192-168-56-141.nip.io/" "Mozilla/5.0 (X11; Linux x86_64; rv:159.0) Gecko/20100101 Firefox/159.0"
192.168.56.1 - - [03/Nov/2026:10:56:28 +0000] "GET /assets/js/skel.min.js HTTP/1.1" 200 3521 "http://estatic
o.192-168-56-141.nip.io/" "Mozilla/5.0 (X11; Linux x86_64; rv:159.0) Gecko/20100101 Firefox/159.0"
192.168.56.1 - - [03/Nov/2026:10:56:28 +0000] "GET /assets/js/util.js HTTP/1.1" 200 3618 "http://estatico.19
2-168-56-141.nip.io/" "Mozilla/5.0 (X11; Linux x86_64; rv:159.0) Gecko/20100101 Firefox/159.0"
dadmin@prod-mtv:~$
```

Exercicio 4. Bloque Directory:

- Captura 1 - Definición do `virtualhost` co bloque `Directory` para desactivar listar o cartafol, activar os enlaces simbólicos e permitir o acceso a tódolos usuarios (paso 2).

```
dadmin@prod-mtv:~$ sudo apache2ctl configtest
sudo systemctl reload apache2
cat /etc/apache2/sites-available/estatico.conf
Syntax OK
<VirtualHost *:80>
    ServerName estatico.192-168-56-141.nip.io
    ServerAlias dimension.192-168-56-141.nip.io portfolio.192-168-56-141.nip.io
    DocumentRoot /var/www/estatico/

    <Directory /var/www/estatico/>
        Options -Indexes +FollowSymLinks
        Require all granted
    </Directory>

    CustomLog /var/log/apache2/estatico-access.log combined
    ErrorLog /var/log/apache2/estatico-error.log
</VirtualHost>
dadmin@prod-mtv:~$
```

- Captura 2 - Mensaxe do navegador na URL `http://portfolio.192-168-56-141.nip.io/images/`, onde xa non se listan os ficheiros (paso 4).

